Preface

This book is a personal adventure. It reflects the present state of an effort on my part to fuse my experience in the medicine of dynamical diseases of the brain with an enduring fascination for the physics that underlies such disease dynamics, and the prospects for controlling such conditions.

The material covered in this book constitutes a one-semester graduate course, *Neural Control Engineering*, that I have taught at Penn State University for the past several years. The students who have taken this course have come from a variety of backgrounds: engineering science, mechanical engineering, aerospace engineering, kinesiology, bioengineering, mathematics, physics, and medicine. To teach such a course, I needed to develop the quantitative science for the students from biology and medicine, and the biology for the quantitative science and engineering students.

There are thirteen chapters in this book. I teach this material by devoting a week to each of the first twelve chapters. I have designed a series of exercises to accompany the first eight chapters. The exercises are designed to develop intuition in formulating computational neuroscience models in a control engineering framework. I have provided some fundamentals of the algorithms that are helpful in crossing the divide from intuition to effective application. I have tested these algorithms in Matlab as well as the open source Octave.

In the remaining weeks of the course, I present lectures on applications covered in chapters 9 through 12, while assigning a series of open problems to the students. The applications of the methods of control engineering presented in this book are, in general, new enough in neuroscience applications that students can produce a final project exploring at the edge of where scientific investigation lies. Some of these projects have reached the quality of publishable advances. Most have, I hope, been of lasting benefit to students who have come to this subject from a broad swath of disciplines.

But perhaps most important for the potential reader, this book reflects the journey of the author, a physician and physiologist by training, to understand this fusion of subjects. I have tried to make the text as self-contained as possible. It should constitute a means to acquire a working knowledge of the fundamentals of control theory and computational neuroscience sufficient to not just understand the literature in this transdisiplinary arena, but

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to begin working near the cutting edge of the research along these lines. It is my hope that this book will serve as a useful guide for the interested scientist, from either the biological or quantitative sciences or engineering, as well as for the interested physician who wishes to gain expertise in these areas.

I owe a special debt to my long-term collaborator and friend, Tim Sauer, who first showed me a copy of the pivotal paper by Voss and colleagues [VTK04] in 2006, and asked me what I thought of it. I suppose in a very real sense, this book is the result of my effort to answer that question. Tim also read the manuscript, and provided invaluable insights and suggestions.

Numerous colleagues have helped guide me toward a working knowledge of control theory. Richard Murray welcomed me to Caltech on several occasions, and his time, wisdom, and advice have been invaluable. My efforts to learn this subject also led me to get to know engineering scholars such as Alok Sinha, Sean Brennan, Jacob (Jack) Langelaan, Constantino Lagoa, Kenneth Jenkins, Raj Acharya, and John Guttag, whose insights into control engineering run deep. I also benefited greatly from attending the lectures on the Mathematical Theory of Control by Alberto Bressan (his book on this topic is superb [BP07]). Three colleagues already working within the interface of neural engineering—Dominique Durand, John White, and Brian Litt—have been of immeasurable help with frequent conversations. I have also, for many years now, benefited from the interest and insightful conversations of Partha Mitra on the potential for fusion of engineering principles with biology.

One of the deep privileges of my life was to have had the chance to encounter, in classrooms and scientific meetings, some of the pivotal people whose work is highlighted in this book. Kenneth Cole, a legendary figure in twentieth-century biophysics, was kind enough to read my first manuscript (with John Moore and Norman Stockbridge), and offer his comments and thoughts—I hope I have represented his work adequately in this book. An entire generation of students at MIT who were enthralled in the class taught by Jerome Y. Lettvin, were inspired to spend the rest of their lives thinking about brains and nervous systems. I benefitted greatly from spending a year studying the physics of the squid axon membrane with John Moore—John is one of the remaining links to the waning era of intense study of axonology. I had a chance to meet John (Jack) Eccles at the Neurosciences Institute in the 1980s, where Vernon Mountcastle, in a long walk on a snowy evening in New York City, convinced me to stay in my neurosurgical residency rather than refocusing my career on science alone. Wilfrid Rall graciously spent time listening to my efforts to better understand reflex variability's role in the spinal cord and cerebral palsy. And scientist-surgeons Arthur Ward, Dennis Spencer, and George Ojemann provided me with much inspiration to seek the science in surgery.

In computational neuroscience, I have had the privilege of long-term friendships with Jack Cowan, G. Bard Ermentrout, Jonathan Rubin, Willliam Troy, Eric Shea-Brown, Giorgio Ascoli, Eugene Izhikevich, Nancy Kopell, Maxim Bazhenov, and John Rinzel, whose collective wisdom has helped guide my professional work.

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I have greatly benefited from my colleagues in Dynamical Systems, including Lou Pecora, Bill Ditto, Eric Kostelich, Edward Ott, Celso Grebogi, James Yorke, Michael Fisher, Predrag Cvitanovic, Mingzhou Ding, Alistair and Moira Steyn-Ross, Andrew Frasier, Rajarshi (Raj) Roy, Marty Golubitsky, Steven Strogatz, Holger Kantz, Thomas Schreiber, Danny Kaplan, Henry Abarbanel, John Beggs, Eshel Ben-Jacob, Raoul Kopelman, Michal Zochowski, Evelyn Sander, Leonard Sander, Michael Unser, Akram Aldroubi, David Collela, John Benedetto, James Collins, Paul Rapp, Frank Moss, Lenny Smith, André Longtin, James Theiler, Mark Spano, Paul So, David Walnut, Arnold Mandell, Alan Garfinkel, Walter Freeman, John Milton, Michael Mackey, Leon Glass, Paul So, Ernest Barreto, and J. Rob Cressman. George Benedek profoundly influenced my view of the possibilities of understanding the physics of biology and medicine—his class notes on biological physics are still available [BV73]. It was also inspirational to quietly sit in the back of H. Eugene Stanley's lectures on biological physics in the 1970s [Sta72]. Harold Morowitz gave me a chance to begin forming a group to fuse physics and neuroscience. Michael Shlessinger helped this physician over more scientific hurdles than I can remember, often over still lawnmowers or tea.

I also am very grateful to Steven Weinstein, an extraordinary children's epileptologist, and to my neuroscience collaborators Karen Gale and Jian-young Wu. A period of productive collaboration with Robert Burke led to a number of discussions of the spinal cord and motoneurons in this text about the brain.

In addition to the blessing of parents who encouraged me to pursue science and medicine, two mentors have had formative influence on my life and work. George Somjen—a deep scholar and superb mentor—taught me how to do science, and much about life and society outside of science. W. Jerry Oakes, the finest physician I have ever met, encouraged me to pursue Pediatric Neurosurgery as well as my scientific pursuits.

Two organizations have given me extraordinary support: the American Association of Neurological Surgeons, which provided a Research Fellowship that helped me pursue completion of my PhD while in the middle of my neurosurgical residency; and the American Physical Society, where my long association with the *Physical Review* has been a rare privilege for a physician.

At Penn State University, David Wormley and Judy Todd saw the value in having a physician join their College of Engineering and the Department of Engineering Science and Mechanics. I hope that I have finally learned enough about engineering to make this gamble seem less of a risk. Jayanth Banavar welcomed me to the Physics Department and provided endless thoughtful advice. Andrew Webb introduced me to the potentials of advanced imaging engineering, which will perhaps find a chapter if this book ever gets to a second edition. The profound support of Robert Harbaugh and Harold Paz is greatly appreciated—they have enabled me to keep my involvement in clinical medicine despite being at continual risk of being consumed by science and engineering. Our scientific institute

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directors at Penn State—Peter Hudson, Susan McHale, and Padma Raghavan—have been endlessly supportive of my efforts to cross disciplinary boundaries.

A special thanks goes to my long-term collaborator, friend, and now fellow "neural engineer" Bruce Gluckman, who helped keep our Center for Neural Engineering afloat while I buried myself in writing this book, and whose brilliant work with electric field control systems will provide a valuable path to implement many of the concepts developed in this book.

This book could never have been worth reading were it not for the students who have suffered through its early materials and drafts in my classes, and offered their suggestions for improvements. I am especially indebted to Andrew Whalen, Jason Mandell, Stephen van Wert, Balaji Shanmugasundaram, Yina Wei, Dana Andre, Guoliang Fang, Tucker Tomlinson, Eric Liauw, Bruce Langford, and Nick Chernyy, and to the long-distance Web attendees Francisco Zabala, John Hobbs, and Óscar Miranda-Dominguez. I extend particular thanks to Patrick Gorzelic, Sergey Gratiy, Madineh Sedigh-Sarvestani, and Nori Okita, for their thoughtful suggestions and corrections to the drafts.

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I would like to warmly acknowledge the generosity of Harvey F. Brush, whose endowment fund supplied the resources necessary to support this adventure in neural control engineering at this early stage.

And finally, I need to express my endless thanks to my wife Eleanor, whose love and tireless support was able to tolerate the seemingly endless amount of time required to write this book.

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